

RUSTOM FIRDOSH BHESANIA

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rustombhesania.github.io/rustom-portfolio

EDUCATION

Relevant Coursework: Digital Signal Processing, Deep Learning for Speech & Text Processing, Microphone Array Processing, Experimental Phonetics, Speech Enhancement and Speech Analysis.

Master of Science in Computational Linguistics, University of Stuttgart, Germany Oct 2024 - Current

Bachelor of Technology in Electronics and Telecommunication Engineering

SVKM's NMIMS UNIVERSITY- Mukesh Patel School of Technology Management and Engineering, Mumbai, India Jul 2016 - Aug 2022

RESEARCH INTERESTS

Speech enhancement, test-time adaptation, ASR, music source separation, music information retrieval. [\[Details\]](#)

ENTREPRENEURSHIP

Cyber Valley AI Incubator (Batch 9) 5 Oct 2026 - 16 Oct 2026
Max Planck Institute for Intelligent Systems *Stuttgart, Germany*

- Building **Voclair**, an on-device, low-latency speech enhancement toolkit and SDK that removes noise and reverberation; pitching at Demo Day on 16 Oct. [\[Details\]](#)

Corporate Campus Challenge (7th Batch), Campus Founders Apr 2026 - Jul 2026
Challenge Partner: Scheer IMC (imc Learning Suite) *Heilbronn, Germany*

- Applied **Design Thinking** to a Scheer IMC challenge, proposing peer-to-peer learning and an AI course recommender for their LMS; presented a working prototype at demo day. [\[Details\]](#)

AWAKE: From Idea to Startup in a Few Weeks, ENI, University of Stuttgart Apr 2026 - Jul 2026

- Developed a karaoke pitch into an “**anywhere karaoke**” concept (two phones as wireless mics, laptop as display) and presented it at the pitch finale. [\[Details\]](#)

EXPERIENCE

Student Assistant, Visualization Research Center (VISUS) May 2026 - Sep 2026
Sedlmair Group, University of Stuttgart *Stuttgart, Germany*

- Built **ViSuS** with Simeon Rau, a rehearsal-recording comparison tool using librosa features, **DTW** and MIDI alignment, and similarity scoring, with a FastAPI backend and two Streamlit clients. [\[Details\]](#)

Student Assistant, Digital Phonetics Lab Apr 2025 - Current
Institute for Natural Language Processing (IMS), University of Stuttgart *Stuttgart, Germany*

- Manage professional audio equipment and the full recording chain, and acquire **high-fidelity audio and music data** in a studio for phonetic and NLP research.

Research Intern Jul 2021 - Apr 2022
K.L. University and Indian Institute of Technology Dharwad, under Dr. K. Samudravijaya *India*

- Evaluated **ASR toolkits** (Kaldi, PyTorch-Kaldi, ESPnet, SpeechBrain) and Wav2Vec; built ASR systems for Assamese, English, Hindi, Marathi and Gujarati, including **end-to-end models** and a custom Parsi (Gujarati) digit database.

ACADEMIC RESEARCH PROJECT

Universal Test-Time Adaptation for Speech Enhancement Dec 2025 - Aug 2026
IMS in collaboration with ISS, University of Stuttgart

Supervisor: Tobias Raichle; Examiner: Dr. Wolfgang Wokurek

Investigated whether **test-time adaptation (TTA)** methods built for denoising transfer to dereverberation, using **CMGAN**. Existing methods degrade on reverberation, but adapting a jointly-trained model on one distortion gives larger gains on the other than adapting on it directly. [\[Details\]](#)

PROJECTS

[\[All projects\]](#)

- WorldSounds (Team Sundarsheng)**

Jul 2026

Carnatic and Jingju knowledge base driving a Markov chain melody generator, built at the [MTG Generative Music AI Workshop](#) (UPF, Barcelona). [\[Details\]](#)
- Karaoke Studio**

Jan 2026

Full-stack app separating vocals from instrumentals with Meta’s Demucs v4 (9.20 dB SDR). [\[Details\]](#)
- Audio Deepfake Detection**

May 2025 - Aug 2025

Identified acoustic features for detecting synthetic speech on ASVspoof 2019; presented at a poster session. [\[Details\]](#)

SKILLS

Technical Skills	Python, Librosa, music21, Opencv, NLTK, pyroomacoustics
Deep Learning Frameworks	Pytorch, Keras
Music Source Separation	Demucs, Spleeter
ASR Tool-kits	Kaldi, pykaldi, pytorch-kaldi, Espnet, Speech Brain
Open Source Tool-kits	Linux, Git, Docker, Bash Scripting